

NIDHI BERDE

nidhiberde1998@gmail.com — linkedin.com/in/nidhi-berde — New York, USA — AWS Certified Solutions Architect – Associate

PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience designing, building and supporting production data pipelines and curated datasets. Proficient in Python, SQL, PySpark, Kafka and Apache Airflow, with hands-on experience in schema design, data quality controls, reconciliation and CI/CD delivery. Track record of executing large-scale on-premises-to-cloud migrations and delivering reliable, well-tested data assets that power analytics, reporting and AI use cases.

TECHNICAL SKILLS

Languages & Query:	Python, SQL (optimization, troubleshooting, analysis), JavaScript
Data Processing:	Apache Spark / PySpark, Kafka (batch & streaming), Apache Airflow, Hive / Hadoop ecosystem, Informatica PowerCenter
Data Formats:	JSON, XML, Parquet, Avro (schema evolution & compatibility)
Platforms & Storage:	AWS (S3, EC2, ELB, DocumentDB), GCP BigQuery, Azure (ADF, Storage, OpenAI), MongoDB, PostgreSQL, MySQL
DevOps & Engineering:	CI/CD, Git, Azure DevOps, Docker, version control & release discipline

PROFESSIONAL EXPERIENCE

Software Engineer – Data Engineering & AI Platform — Tata Consultancy Services, Edison, NJ *June 2024 – Present*

- **Customer Deployment & Production Agents:** Built and deployed a production-grade AI agent platform tailored to an aerospace & defense customer — owning end-to-end delivery from requirements through production, processing 10,000+ documents across 15+ file types on AWS (EC2, S3, ELB) with Docker and automated CI/CD.
- **Agent Architecture & Prompt Engineering:** Built a Python and FastAPI-based agent service — designing prompts, tool/function calls and reasoning flows for NLP-driven classification and autonomous SQL execution, iterating on agent behaviour to improve reliability and output quality for downstream consumers.
- **Iterative Agent Improvement:** Enhanced the production AI agent pipeline with image extraction capability — identified a per-image API call bottleneck, redesigned the approach to consolidate all images per document into a single agent call, and measured improvement in end-to-end throughput.
- **Agent Quality Measurement & Observability:** Implemented NLP-driven validation controls and a structured quality measurement layer across 10,000+ documents — persisting results in MongoDB for full lifecycle observability and traceability, enabling root-cause analysis and continuous improvement of agent output quality.

Data Engineering Intern — Tata Consultancy Services, Edison, NJ *June 2023 – August 2023*

- Built end-to-end ETL pipelines in Python using Informatica PowerCenter, integrating Oracle, SQL Server and Excel sources into cloud platforms; applied validation, cleansing and transformation to ensure data integrity across distributed back-end systems.

Application Development Associate — Accenture, Mumbai, India *January 2021 – August 2022*

- **Enterprise Customer Deployment:** Partnered directly with a global FMCG client to execute a 30+ TB migration from on-premises Hadoop to GCP BigQuery — converting HiveQL to BigQuery SQL, managing schema evolution and applying partitioning strategies that achieved a 35% query performance improvement.
- **Domain Data Layer — Supply Chain:** Designed 15+ curated BigQuery tables (raw, refined and aggregated layers) tailored to the client's supply chain workflows; conducted rigorous reconciliation and integrity checks across 30+ datasets to validate the solution before handoff.

PROJECTS

- **XML-to-Parquet Data Pipeline:** Built a Python-based batch pipeline transforming XML configuration data into structured Parquet datasets with schema validation and lineage traceability — improving data usability by 80%.
- **Agentic AI Pipeline:** Designed and built an end-to-end agentic AI application (Python, FastAPI, SQLAlchemy, pandas) — engineering prompts, tool/function calls and reasoning flows for a large language model (Groq, llama-3.3-70b) to autonomously ingest, classify and synthesise insights from live data sources; iterated on prompt design and agent behaviour to improve output reliability.

EDUCATION

M.S. Computer Science (2024) — Pace University, New York, USA — GPA: 3.8

B.E. Computer Engineering (2020) — University of Mumbai, India

CERTIFICATIONS

AWS Certified Cloud Practitioner, AWS Certified Solutions Architect – Associate